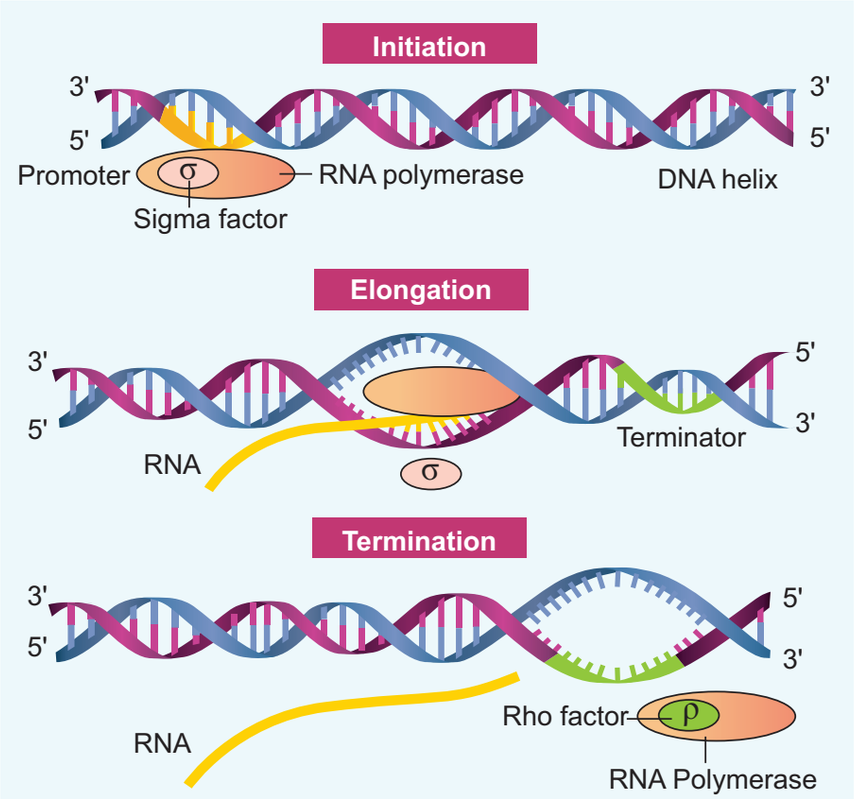


Fig. 5. 8 Process of transcription in prokaryotes

polymerase is only capable of catalyzing the process of elongation. The RNA polymerase associates transiently with initiation factor sigma (σ) and termination factor rho (ρ) to initiate and terminate the transcription, respectively. Association of RNA with these factors instructs the RNA polymerase either to initiate or terminate the process of transcription (**Fig. 5.8**).

In bacteria, since the mRNA does not require any processing to become active and also since transcription and translation take place simultaneously in the same compartment (since there is no separation of cytosol and nucleus in bacteria), many times the translation can begin much before the mRNA is fully transcribed. This is because the genetic material is not separated from other cell organelles by a nuclear membrane consequently; transcription and translation can be coupled in bacteria.

In Eukaryotes, there are at least three RNA polymerases in the nucleus (in addition to RNA polymerase found in the organelles). There is a clear division of labour. The RNA polymerase I transcribes rRNAs (28S, 18S and 5.8S), whereas the RNA polymerase III is responsible for transcription of tRNA, 5S rRNA and snRNA. The RNA polymerase II transcribes precursor of mRNA, the hnRNA (heterogenous nuclear RNA). In eukaryotes, the monocistronic structural genes have interrupted coding sequences known as **exons** (expressed sequences) and non- coding sequences called **introns** (intervening sequences). The introns are removed by a process called **splicing**. hnRNA undergoes additional processing called **capping** and **tailing**. In capping an unusual nucleotide, methyl guanosine triphosphate is added at the 5' end, whereas adenylate residues (200-300) (Poly A) are added at the 3' end in tailing (**Fig. 5.9**). Thereafter, this processed hnRNA, now called mRNA is transported out of the nucleus for translation.

The split gene feature of eukaryotic genes is almost entirely absent in prokaryotes. Originally each exon may have coded for a single polypeptide chain with a specific function. Since exon arrangement and intron removal are flexible, the exon coding for these polypeptide subunits act as domains combining in various ways to form new genes. Single genes can produce different functional proteins by arranging their exons in several different ways through alternate splicing

XII Std Biology-Zoology Chapter-5 EM.indd 70 23/12/2022 12:15:59